

Summary Report

1. Techniques Used

- Data was collected from multiple sources including **CSV, JSON, SQL (SQLite), and REST API**
 - All datasets were merged using common keys: **customer_id** and **product_id**
 - Performed **Exploratory Data Analysis (EDA)** using:
 - Histograms
 - Heatmaps
 - Scatter plots
 - Pairplots
 - Grouped statistical analysis
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2. Biggest Issues with Raw Data

- Missing values observed in **age** and **income** (approximately 5%)
 - Lack of datetime features — derived:
 - days_since_signup
 - days_since_last_purchase
 - Mixed data types in **customer_id**, resolved using regex-based cleaning
 - No predefined target variable — created **purchased flag** from transaction records
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3. Imputation Comparison

Method	Rows Lost	Remarks
Simple Imputer	0	Fast but ignores feature relationships
Random Sample	0	Maintains original data distribution
KNN Imputer	0	Captures similarity between observations

Method	Rows Lost	Remarks
MICE	0	Most accurate but computationally expensive
CCA	~50	Results in data loss — not preferred

Conclusion:

KNN and MICE imputation methods performed best in preserving relationships between features.

4. Feature Scaling Evaluation

- **RobustScaler** performed best for skewed data with outliers (e.g., income, transaction amount)
 - **StandardScaler** was effective for normally distributed features such as age
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5. Final Dataset

- All missing values handled successfully
 - All categorical variables encoded appropriately
 - New meaningful features engineered
 - Final processed dataset exported as **processed_customer_data.csv**
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